

Project Summary

Exhibit A

Project Title: Urological Procedure Digital Twin

Sponsor: Boston Scientific

Department: Biomedical Engineering

Desired Team Composition (disciplines): Biomedical Engineering

Start Date: [Click here to enter a date.](#)

End Date: [Click here to enter a date.](#)

Sponsorship Fee: enter \$ amount.
type.

Sponsorship Type: select

Background & Overview

Boston Scientific (BSC) is dedicated to transforming lives through innovative medical solutions that improve the health of patients around the world. Founded in 1979 by John Abele and Pete Nicholas, the company now employs approximately 59,000 people worldwide and offers a portfolio of products that treat more than 48 million patients annually. BSC's Urology division provides a comprehensive range of technologies for treating kidney stones using ureteroscopy, a minimally invasive procedure. During ureteroscopy, a ureteroscope is inserted through the urethra, bladder, and ureter into the kidney. The ureteroscope is equipped with a camera, and saline irrigation is delivered through the scope to maintain a clear field of view. Devices used during the procedure influence irrigation flow rates, which in turn affect intrarenal pressure and procedural visualization.

Problem/Opportunity Statement

New urological devices are traditionally evaluated using animal testing to assess procedural performance and visualize the surgical field under clinically relevant conditions. However, ethical considerations, increasing regulatory support for alternatives to animal testing, and the need for more scalable evaluation methods have created an opportunity to develop computational approaches for device assessment. Boston Scientific has collected synchronized ureteroscopy videos, kidney pressure measurements, and irrigation flow rate data from experimental studies. The goal of this project is to develop machine learning models that characterize the relationships between endoscopic imagery,

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pressure, and flow over time. These models will serve as the foundation for a digital twin capable of generating realistic synthetic ureteroscopy images or video from specified pressure and flow profiles, ultimately supporting product development while reducing reliance on animal testing.

Project Significance

Boston Scientific is committed to advancing innovative product development while reducing reliance on animal testing through the adoption of computational modeling and artificial intelligence. This project supports that objective by developing machine learning (ML) models that capture the relationships between ureteroscopic imagery, irrigation flow, and kidney pressure. A successful proof of concept would establish the foundation for a digital twin capable of generating realistic synthetic procedural data, enabling faster device evaluation, supporting evolving regulatory expectations, and reducing the cost, time, and ethical burden associated with traditional preclinical testing.

Illustration, Drawing, Image, or Graphic Describing the Project

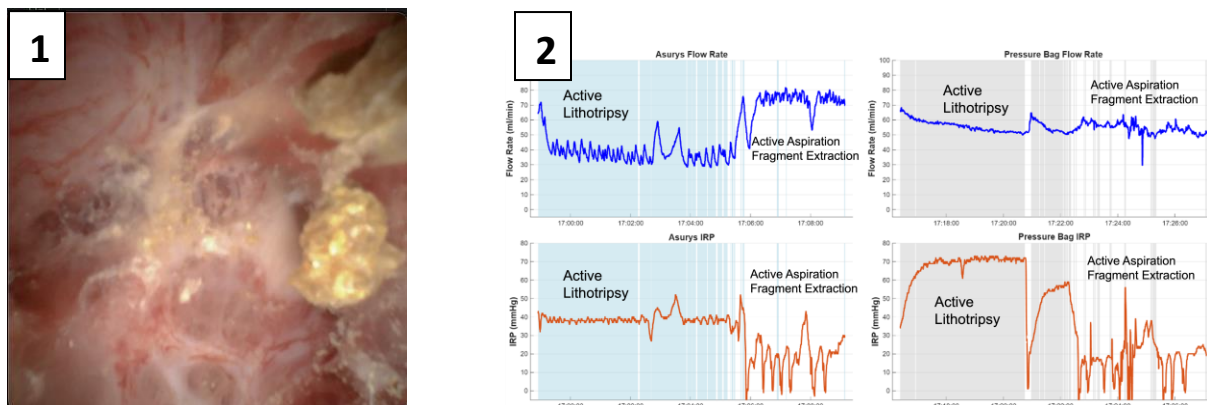


Figure 1: Endoscopic view of a renal calculus during ureteroscopy in a porcine model.

Figure 2: Head-to-head preclinical case study of Physician 1 comparable stones treated (each canine cystine 0.3g/0.5-1cm) with resulting plot of IRP or flow rate from irrigation with either Asurys™ (left) or pressure bag (right). Shading indicates when laser lithotripsy was active.

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Anticipated Outcomes of the Student Team

Project Definition: Work with Boston Scientific to define project objectives, success metrics, data requirements, and validation criteria. Establish performance metrics for image quality, accuracy, and computational efficiency. Identify project constraints including data availability, model interpretability, and regulatory considerations.

Background Research: Conduct a literature review on ureteroscopy, kidney irrigation dynamics, synthetic data generation, computer vision, multimodal learning, and generative AI/ML techniques. Become familiar with Boston Scientific's current testing methodology and FDA initiatives promoting alternatives to animal testing.

Data Exploration & Preparation: Analyze the provided ureteroscopy video, kidney pressure, and flow rate datasets. Synchronize multimodal data streams, perform preprocessing and feature extraction, identify data quality issues, and develop a pipeline for efficient model training and evaluation.

Model Development: Design, implement, and compare machine learning models that learn the relationships between endoscopic imagery, irrigation flow rate, and kidney pressure over time. Evaluate multiple architectures for predictive modeling and synthetic image/video generation, selecting the most effective approach using quantitative performance metrics.

Model Validation & Proof of Concept: Validate model performance using held-out datasets and compare generated images against real procedural videos using image similarity metrics, temporal consistency, and domain-specific performance measures. Assess the model's ability to generate realistic visualizations from specified pressure and flow profiles and document strengths, limitations, and opportunities for future improvement.

Documentation and Presentation: Produce well-documented code, reproducible training and evaluation workflows, a comprehensive final report describing methodology and results, and presentations for both Boston Scientific and the Enterprise Design Expo. Deliver recommendations for future development and potential integration into Boston Scientific's product evaluation workflow.

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Key Functional Requirements of the Material, Product, Process, or System

The system shall:

- Ingest synchronized ureteroscopy video, kidney pressure, and irrigation flow rate data.
- Align multimodal data streams with sufficient temporal accuracy for model training.
- Develop predictive models that quantify the relationship between visual appearance, pressure, and flow.
- Generate synthetic ureteroscopy images and/or video sequences from specified pressure and flow profiles.
- Evaluate generated images using quantitative image quality metrics and comparison with real procedural data.
- Provide a machine learning framework that can incorporate additional datasets as they become available.
- Produce documentation, code, and validation results that can be transferred to Boston Scientific for future development.

Special Notes

- Students will gain experience applying modern AI and machine learning techniques, including computer vision, time-series modeling, and generative models.
- Strong background in programming and data analysis is required.
- Experience with image processing and machine learning techniques is beneficial.
- Previous exposure to the medical device industry is beneficial.
- The project aligns with current industry trends toward reducing animal testing through computational modeling and synthetic data generation.

Sponsor Contact Information

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Sponsor to Provide

- Sponsorship funding for the BME Senior Design program and the project team.
- Regular interaction with the student team via tele/video conference (suggest weekly). Opportunities for face-to-face interactions are encouraged, both on campus and at Boston Scientific at sponsor discretion.
- Boston Scientific will provide de-identified experimental datasets and technical guidance throughout the project.
- Boston Scientific will provide funding for software development tools required for project completion.

Michigan Tech to Provide

- An undergraduate project team of senior level students majoring in Biomedical Engineering.
- Faculty advising support from the Biomedical Engineering Department.

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- Team access to key department facilities and resources for design, proof of concept, and evaluation.
- Team access to a portion of the sponsorship fee for direct project expenses.
- A comprehensive final report, presentation, and poster display at Michigan Tech's Design Expo.

Reference 1: <https://www.bostonscientific.com/en-US/about-us.html>